

Programme	B. Sc. Computer Science				
Course Title	Programming fundamentals using C				
Type of Course	Minor				
Semester	III				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Basic Computer Literacy 2. Basic Problem-Solving Skills				
Course Summary	This course teaches the basics of programming using the C language. C is a powerful and widely used programming language known for its efficiency and flexibility. Through this course, students will learn how to write, understand and debug C code to solve various problems and build simple applications.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate a solid understanding of fundamental programming concepts	An	P	Instructor-created lab exams / Quiz
CO2	Develop effective problem-solving skills by applying algorithmic thinking and logical reasoning.	An	P	Problem-solving assessments
CO3	Gain proficiency in writing, compiling, debugging, and executing	Ap	P	Modelling Assignments

	C programs to implement algorithms, solve problems, and create applications.			
CO4	Learn techniques to write efficient and optimized C code, including memory management, algorithm design, and performance tuning, to produce high-quality and scalable software solutions.	Ap	P	Modelling Assignments/ / Case studies
CO5	Understand and apply software development practices such as modular programming, code documentation and debugging techniques to write maintainable and robust C programs.	Ap	P	Modelling Assignments/ / Case studies
CO6	Develop critical thinking skills by analyzing and evaluating C code, identifying errors and inefficiencies, and proposing solutions to improve code quality and performance.	Ap	P	Hands-on exercises
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I		Problem solving and logical Thinking	10	15
	1	Overview of computational thinking concepts. Definition of algorithm and its characteristics .Understanding the importance	2	

		of algorithms in problem-solving		
	2	Algorithm Development:Steps involved in designing algorithms	2	
	3	Pseudocode as an intermediate step in algorithm development.	1	
	4	Understanding flowchart symbols and their meanings .Learning to represent algorithms using flowcharts.	2	
	5	Raptor as a precursor to text-based programming languages	2	
	6	Drawing simple flowcharts	1	
II	Introduction to C		10	20
	7	Structure of C program	2	
	8	C Character Set, Keywords, Identifiers	1	
	9	Data Types, Variables, Declarations, Symbolic Constants	2	
	10	Operators :Arithmetic, Logical, Relational & Equality, and Unary, Operator Precedence and Associativity	2	
	11	Library Functions, Comments	1	
	12	I/O functions- Formatted scanf() & printf().	2	
III	Control Statements,Arrays & Strings		14	20
	13	Selection Statements: if, if-else, switch	3	
	14	Iteration: while, do while, for	4	
	15	Arrays: One dimensional and Two Dimensional(introduction only)	3	
	16	Strings:Basic string handling functions	2	
	17	Structure :Definition,Processing-period Operator, Union(Concepts only)	2	
1V	User defined Functions		11	15
	18	Definition of function,Advantages, Understanding function prototypes and declarations	3	
	19	Introduction to function definitions and function calls	3	
	20	Exploring function parameters : Actual and Formal parameters	2	
	21	Recursion	2	

	22	Pointers-declarations(Basic concept only)	1	
V	Hands-on C:		30	
	Practical Applications, Case Study and Course Project			
	1	Write a C program using Variables and Data Types Write a C program using Arithmetic Operations Write a C program using Loops Write a C program using Arrays Write a C program using Functions Write a C program using Strings	20	
	2	Case study: 1. Library Management System: 2. Develop a program to manage a library's collection of books. Implement functions for adding, removing, and searching for books. 3. Ticket Booking System: Design a program to manage ticket bookings for a cinema or theater.	5	
	3	Capstone/Course Project: Design a real-time project in C	5	

Reference:

1. Balagurusamy, E. Programming in ANSI C. Tata McGraw-Hill Education, 2019.
2. King, K. N. C Programming: A Modern Approach. 2nd ed., W. W. Norton & Company, 2008.
3. Kernighan, Brian W., and Dennis M. Ritchie. The C Programming Language. 2nd ed., Prentice Hall, 1988.
4. Prata, Stephen. C Primer Plus. 6th ed., Addison-Wesley, 2013.
5. Perry, Greg. Absolute Beginner's Guide to C. 3rd ed., Que Publishing, 2014.
6. Oualline, Steve. Practical C Programming. 3rd ed., O'Reilly Media, 1997.
7. Hanly, Jeri R., and Elliot B. Koffman. Problem Solving and Program Design in C. 8th ed., Pearson, 2016.
8. Gottfried, Byron S. Programming with C. 2nd ed., McGraw-Hill, 1996.
9. Holmes, Dan. C in a Nutshell. 2nd ed., O'Reilly Media, 2015.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PS O5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	1	-	-	-	1						
CO 2	1	-	2	-	-	-						

CO 3	-	-	2	-	-	-						
CO 4	-	1	3	3	-	3						
CO 5	-	2	3	3	-	3						
CO 6	-	-	-	-	-	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1		✓		✓
CO 2	✓	✓		✓
CO 3		✓		✓
CO 4	✓			✓
CO 5	✓		✓	✓
CO 6	✓		✓	✓